

TAPICIN POWDER FOR INJECTION 4.5G



Ingredient(s):

Piperacillin (as Sodium) 4.0g Tazobactam (as Sodium) 0.5g

Pharmacodynamic:

1. Piperacillin is a broad spectrum, semi synthetic penicillin active against many gram-positive and gram-negative aerobic and anaerobic bacteria. It exerts bactericidal activity by inhibition of both septum and cell wall synthesis.
2. Tazobactam, a triazolyl-methyl penicillanic acid sulfone, is a potent inhibitor of many β -lactamases, in particular the plasmid mediated enzymes which commonly cause resistance to penicillins and cephalosporins including the 3rd generation cephalosporins.
3. The presence of Tazobactam in Tapicin enhances and extends the antibiotic spectrum of piperacillin to include many β -lactamases-producing bacteria normally resistant to it and to other β -lactam antibiotics. This combination is highly active against piperacillin-sensitive microorganisms as well as β -lactamase-producing, piperacilline-resistant microorganisms:

Gram-Negative Bacteria: Most plasmid mediated β -lactamase- and non- β -lactamase-producing strains of *E.coli* and *Klebsiella* spp., (including *K.oxytoca*, *K.pneumoniae*), *Proteus* sp. (including *P.vulgaris*, *P.mirabilis*), *Salmonella* and *Shigella* spp., *Neisseria gonorrhoeae*, *Neisseria meningitidis*, *Moraxella* sp. (including *M. catarrhalis*), *Haemophilus* sp. (including *H.influenzae*, *H.parainfluenzae*), *Pasteurella multocida*, *Yersinia* and *Campylobacter* spp., *Gardnerella vaginalis*. Many chromosomally-mediated β -lactamase-and non- β -lactamase producing strains of *Enterobacter* sp. (including *E.cloacea*, *E.aerogenes*), *Citrobacter* sp. (including *C.freundii*, *C. diversus*), *Providencia* sp., *Morganella morganii*, *Serratia* sp. (including *S.marcescens*, *S. Liquefaciens*), *Pseudomonas aeruginosa* and other *Pseudomonas* sp. (including *P.cepacia*, *P.fluorescens*), *Xanthomonas maltophilia*, *Acinetobacter* sp.

Gram-Positive Bacteria: β -lactamase-and non- β -lactamase-producing strains of streptococci (*S.pneumoniae*, *S.pyogenes*, *S.bovis*, *S.agalactiae*, *S.viridans*, Groups C and G), enterococci (*E.faecalis*), *Staphylococcus aureus* (not methicillin-resistant *S.aureus*), *S.saprophyticus*, *S.epidermidis* (coagulase-negative staphylococci), *corynebacteria*, *Listeria monocytogenes*, *Nocardia* sp.

Anaerobic Bacteria: β -lactamase- and non β -lactamase-producing anaerobes, eg *Bacteroides* sp. (including *B.bivius*, *B. disiens*, *B.capillosus*, *B.melaninogenicus*, *B.oralis*), the *Bacteroides fragilis* group (including *B.fragilis*, *B.vulgatus*, *B.distasonis*, *B.ovatus*, *B.thetaiotaomicron*, *B.uniformis*, *B.asaccharolyticus*) as well as, *Peptostreptococcus* and *Fusobacterium* spp., *Eubacterium* group, *Clostridia* sp. (including *C.difficile*, *C.perfringens*), *Veillonella* and *Actinomyces* spp.

Pharmacokinetic:

1. Peak plasma concentrations of piperacillin and tazobactam are attained immediately after completion of an IV infusion of this product.
2. The plasma half-life of piperacillin and of tazobactam ranged from 0.7 – 1.2 hours and was unaffected by dose or duration of infusion.
3. Piperacillin is metabolized to a minor microbiologically active desethyl metabolite, while tazobactam to a single metabolite that lacks pharmacological and antibacterial activities.
4. Both piperacillin and tazobactam are eliminated via the kidney by glomerular filtration and tubular secretion.
5. Piperacillin is excreted rapidly as unchanged drug with 68% of the administered dose excreted in urine.
6. Tazobactam and its metabolite are eliminated primarily by renal excretion with 80% of the administered dose excreted as unchanged drug and the remainder as the single metabolite.
7. Piperacillin, tazobactam and desethyl piperacillin are also excreted into the bile.
8. Piperacillin and tazobactam are widely distributed into tissues and body fluids including intestinal mucosa, gallbladder, lung, female reproductive tissues (uterus, ovary and fallopian tube), interstitial fluid and bile.
9. Mean tissue concentrations are generally 50–100% of those in plasma.
10. Half-life for piperacillin and tazobactam increases with decreasing creatinine clearance in renal impairment (kindly refer to recommended dose for dosage adjustment).
11. Hemodialysis removes 30–40% of piperacillin/tazobactam dose with an additional 5% of the tazobactam dose removed as the tazobactam metabolite.
12. Peritoneal dialysis removes approximately 6 and 21% of the piperacillin and tazobactam doses, respectively, with up to 16% of the tazobactam dose removed as the tazobactam metabolite.

Indication(s):

1. Treatment of the following systemic and/or local bacterial infections in which susceptible organisms have been detected or suspected, lower respiratory tract infection, urinary tract infections (complicated and uncomplicated), intra-abdominal infections, skin and skin structure infections, and bacterial septicaemia.
2. Polymicrobial infections including those where aerobic and anaerobic organisms are suspected (intra-abdominal, skin and skin structure infection, lower respiratory tract).
3. In Combination with aminoglycoside is indicated for bacterial infections in neutropenic adults or children.

Dosage and Administration:

1. **Adults and children > 12 years with normal renal function:**

Usual dosage: 4.5g every 8 hours.

The total daily dosage depends on the severity and localization of the infection and can vary from 2.25- 4.5g administered every 6 or 8 hours.

2. **Neutropenia:** Recommended dose: 4.5g every 6 hours in combination with an aminoglycoside. Neutropenic patients with signs of infection (eg. Fever) should receive immediate empirical antibiotic therapy before laboratory results are available.
3. **Children < 12 years:** This product is only recommended for the treatment of children with neutropenia.

Children weighing >50kg: Follow adult dosing guidance. Including the aminoglycoside.

Children with normal renal function and weighing <50kg: Adjust the dose to 90mg/kg (80mg piperacillin / 10mg tazobactam) administered every 6 hours in combination with an aminoglycoside.

4. **Elderly:** This product may be used at the same dose levels as adults except in cases of renal impairment.

5. **Renal Insufficiency:**

- a) **Adults and children weighing > 50kg:** The IV dose should be adjusted to the degree of actual renal impairment. The suggested daily doses are as follows:

Creatinine Clearance	Recommended Piperacillin / Tazobactam Dosage
20-80ml/min	12g/1.5g/day in divided doses of 4g/500mg every 8 hours.
<20ml/min	8g/1g/day in divided doses of 4g/500mg every 12hours.

Patients on haemodialysis: The maximum daily dose is 8g/1g. Because haemodialysis removes 30-50% of piperacillin in 4 hours, 1 additional dose of 2g/250mg should be administered following each dialysis period.

Patient with renal failure and hepatic insufficiency: Measurement of serum levels of Piperacillin/Tazobactam will provide additional guidance for adjusting dosage.

- b) **Adults and Children weighing <50kg:** The IV dosage should be adjusted to the degree of actual renal impairment as follows:

Creatinine Clearance	Recommended Piperacillin / Tazobactam Dosage
40-80	90mg (80mg piperacillin / 10mg tazobactam)/kg every 6 hours
20-40	90mg (80mg piperacillin / 10mg tazobactam)/kg every 8 hours
<20	90mg (80mg piperacillin / 10mg tazobactam)/kg every 12hours

Children <50kg on haemodialysis: The recommended dose is 45mg/kg every 8 hours. Each patient must be monitored closely and dosage regimen to be adjusted accordingly.

Duration of therapy: Therapy is recommended to be a minimum of 5 days and maximum of 14 days, considering that dose administration should continue at least 48 hours after the resolution of clinical signs and symptoms or fever.

Mode of Administration

This product may be given by slow IV injection (3-5 minutes) or by infusion (20-30minutes).

Reconstitution:

IV injection: Each vial of 4.5g should be reconstituted with 20ml of one of the reconstitution diluent respectively. Swirl until dissolved.

IV infusion: Each vial should be reconstituted with at least 20ml of one of the reconstitution diluents. The reconstituted solution should be further diluted to the desired volume (50-150ml) with one of the reconstitution diluents.

Reconstitution diluents: Water for injection (max volume used for each dose is 50ml), 5% Dextrose injection, 0.9% Sodium Chloride injection.

This product do not contain preservative, therefore appropriate aseptic technique should be performed during preparation. Use the reconstituted solution immediately after preparation.

Contraindications:

Patients with a history of allergic reactions to any of the penicillins and/or cephalosporins or β -lactamase inhibitors.

Precaution(s)/Warning(s):

1. Serious and occasionally fatal anaphylactic reactions have been reported in patients receiving therapy with penicillins. If allergic reaction occurs during therapy with this product, the antibiotic should be discontinued. Serious hypersensitivity reactions may require adrenaline and other emergency measures.
2. Periodic assessment of organ system functions including renal, hepatic and hematopoietic during prolonged therapy is advisable.
3. Bleeding manifestations have occur in some patients receiving β -lactam antibiotics. This may associated with abnormalities of coagulations tests. If bleeding manifestations occur, the product should be discontinued and appropriate therapy instituted.
4. In case of severe, persistent diarrhea, the possibility of antibiotic-induced, life-threatening pseudomembranous colitis must be taken into consideration. Therapy must be discontinued immediately and suitable therapy for *Clostridium difficile* colitis (e.g: Metronidazole or vancomycin) to be initiated. Preparations which inhibit peristalsis are contraindicated.
5. During prolonged treatment, microbiological follow-up maybe required to detect any important superinfections.
6. Patients may experience neuromuscular excitability or convulsions if higher than recommended IV doses are given.
7. Periodic electrolyte determinations should be made in patients with low potassium reserves. These patients may have possibility of hypokalaemia and modest elevation of indices of liver function.
8. Antimicrobials used in high doses for short periods to treat gonorrhoea may mask or delay the symptoms of incubating syphilis, so prior to treatment, patients with gonorrhoea should also be evaluated for syphilis.
9. Rare cases of haemophagocytic lymphohistiocytosis (HLH) have been observed following therapy (> 10 days) with piperacillin, often as a complication of DRESS. HLH is a pathologic immune activation which leads to excessive systemic inflammation and can be life-threatening and early diagnosis and rapid initiation of immunosuppressive therapy is essential. Characteristic signs and symptoms include fever, hepatosplenomegaly, cytopenias, hyperferritinaemia, hypertriglyceridaemia, hypofibrinogenaemia, and haemophagocytosis. If piperacillin is suspected as possible trigger, treatment should be discontinued.

Interaction with Other Medicaments

1. Concurrent administration of probenecid and piperacillin/tazobactam produced a longer half-life and lower renal clearance for both piperacillin and tazobactam.
2. Whenever this product is used with another antibiotic, especially aminoglycoside , the drug must not be mixed in intravenous solutions or administered concurrently due to physical incompatibility.
3. During simultaneous administration of high doses of heparin, oral anticoagulants and other drugs which may affect the blood coagulation system and/or the thrombocyte function, the coagulation parameters should be tested more frequently and monitored regularly.
4. Piperacillin has been reported to prolong the action of vecuronium. Caution should be exercise when used perioperatively with vecuronium and similar neuromuscular-blocking agents.
5. Penicillin may reduce the excretion of methotrexate. Serum level of methotrexate should be monitored in patients on high-dose methotrexate therapy.
6. Caution for usage:
 - a) This product should not be mixed with other drugs in syringe or infusion bottle since compatibility has not been established.
 - b) Whenever this product is used concurrently with another antibiotic, the drug must be administered separately.
 - c) This product should not be used with solutions containing only sodium bicarbonate.
 - d) This product should not be added to blood products or albumin hydrolysates.

Pregnancy and Lactation

Pregnant women should be treated only if the expected benefit overweighs the possible risks to the pregnant women and foetus. Caution should be exercised when this product is administered to a nursing woman. Piperacillin is excreted in low concentrations in human milk, Tazobactam concentrations in human milk have not been studied.

Side Effect(s)/Adverse Reaction(s):

1. Rarely, significant leucopenia may be associated with prolonged therapy. Very rarely, interstitial nephritis may occur.
2. Hepatitis and cholestatic jaundice have been reported rarely with some penicillins and β -lactamase inhibitors.
3. Haemolytic anemia has been reported rarely with penicillin and piperacillin/tazobactam.
4. Adverse local reactions that were reported were phlebitis and thrombophlebitis.
5. Systemic adverse clinical reactions reported were diarrhea, rash, erythema, pruritus, vomiting, allergic reactions, nausea, urticaria, superinfections, skin reactions, eruption, increased sweating, erythema, multiforme, eczema, exanthema, maculopapular rash, soft/loose stools, stomatitis, constipation, muscular weakness, hallucination, dry mouth, hypotension, muscle pain, superficial phlebitis, fever, hot flushes, oedema, tiredness.
6. Local reactions included injection site inflammation and injection site pain when solution was not prepared according to the recommendation.
7. Adverse laboratory changes without regard to drug relationship that were reported were transient reduction in the white blood cell count (leucopenia), eosinophilia, thrombocytosis, thrombocytopenia, positive Coombs' test, hypokalemia, transient rise in the serum levels of liver enzymes, bilirubin and increased level of renal function parameters have been detected in serum.

Symptoms and Treatment for Overdosage, and Antidote(s):

No specific antidote is known. In case of motor excitability or convulsions, anticonvulsive agents may be indicated. In the case of severe, anaphylactic reactions, the usual countermeasures are to be initiated.

Excessive serum concentrations of piperacillin/tazobactam may be reduced by hemodialysis.

The majority of overdose events experienced, including nausea, vomiting, and diarrhea, have also been reported with the usual recommended dosages. Patients may experience neuromuscular excitability or convulsions if higher than recommended doses are given intravenously (particularly in the presence of renal failure).

Shelf life before open:

2 years

Shelf life after reconstitution:

24 hours at 25°C

7 days under refrigeration (2°C - 8°C)

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

For reconstituted solution:

This product is compatible with one of following reconstitution diluents: Water for Injection (WFI), 5% Dextrose, 0.9% Sodium chloride injection. These solutions are stable up to 24 hours at 25°C or 7 days under refrigeration (2°C - 8°C).

Product Description and Packing(s):

Before reconstitution: A white or almost white powder in vial

After reconstitution: Colorless to slightly yellow solution

4.5g x 10 vials / Box



Manufacturer:

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